

Using the Hioki Natural Language to SQL (NL2SQL) application

Quick Start

Note: all images used in this file are referenced from the PowerPoint also in this directory. For an abbreviated guide, use the PowerPoint.

As of 2026-05-21, the program lives at P:/PE III/Personal/Eric/Projects/+HiokiInterfaceExecutable.



Running the program

Name	Date modified	Type	Size
wwwroot	4/29/2026 12:03 PM	File folder	
+loadDbEnv.bat	4/29/2026 12:23 PM	Windows Batch File	1 KB
+README.txt	5/21/2026 8:41 AM	Text Document	1 KB
+UserGuide.pptx	5/21/2026 8:25 AM	Microsoft PowerP...	3,896 KB
+UserGuideExtended.pdf	3/2/2026 2:36 PM	Adobe Acrobat D...	2,451 KB
appsettings.Development.json	4/29/2026 11:27 AM	JSON Source File	1 KB
appsettings.json	4/29/2026 11:26 AM	JSON Source File	1 KB
aspnetcorev2_inprocess.dll	4/17/2026 4:04 PM	Application exten...	384 KB
HiokiNL2SQLMark1.exe	5/21/2026 7:31 AM	Application	116,191 KB
HiokiNL2SQLMark1.pdb	5/21/2026 7:31 AM	Program Debug D...	108 KB
HiokiNL2SQLMark1.staticwebassets.endp...	4/29/2026 11:39 AM	JSON Source File	69 KB
InterProcessIO.pdb	5/21/2026 7:31 AM	Program Debug D...	17 KB
Microsoft.Data.SqlClient.SNI.dll	1/15/2025 7:23 PM	Application exten...	554 KB
web.config	5/21/2026 7:31 AM	Configuration Sou...	1 KB

The folder with the program resides at the path P:/PE III/Personal/Eric/Projects/+HiokiInterfaceExecutable as of 2026-05-21. It should look something like this. Files without a '+' are for the program and should not be touched.

Before you run the .exe, make sure to run +loadDbEnv.bat (or just loadDbEnv if you hide file extensions) first, otherwise you'll just see the following:

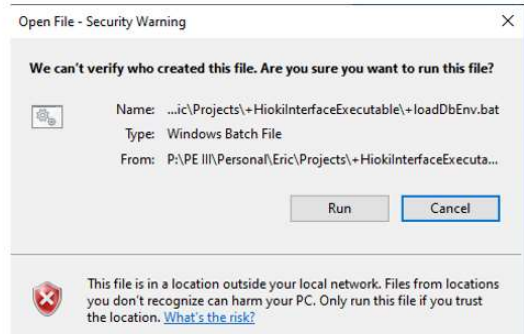




The .bat file

Name	Date modified	Type	Size
wwwroot	4/29/2026 12:03 PM	File folder	
loadDbEnv.bat	4/29/2026 12:23 PM	Windows Batch File	1 KB
README.txt	5/21/2026 8:41 AM	Text Document	1 KB
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Microsoft.Data.SqlClient.SNI.dll	1/15/2025 7:23 PM	Application Extension	554 KB
web.config	5/21/2026 7:31 AM	Configuration Source File	1 KB

The .bat file loads a few environment variables to offload database credentials from the app. Because it automatically executes commands and doesn't live on your computer's hard drive, it's flagged as malicious even if you've run it before. By default, double clicking this will execute its contents straight from the terminal. If you want to see what it's doing, you're welcome to open it in a text editor by right-clicking, then selecting edit (although you will still need to confirm the security dialog). Doing so will not execute the commands.



What you'll see if you run straight from the P drive

When you run the .bat file, you'll see a terminal window pop up briefly, then disappear. That means the file has run and you're ready to run the actual program.

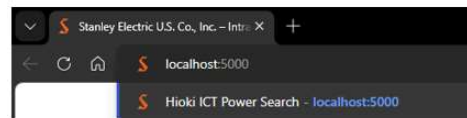


The .exe file

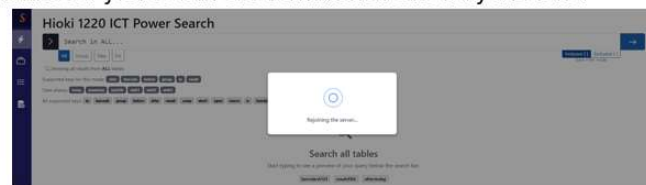
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Microsoft.Data.SqlClient.SNI.dll	1/15/2025 7:23 PM	Application Extension	554 KB
web.config	5/21/2026 7:31 AM	Configuration Source File	1 KB

Once you've run the .bat file, you're ready to go. Double clicking the .exe file will show the security dialog (because it's an executable), but after you confirm, it should show something like this:

```
Microsoft.Hosting.Lifetime[14]
info: Now listening on: http://localhost:5000
info: Microsoft.Hosting.Lifetime[0]
info: Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
info: Hosting environment: Production
info: Microsoft.Hosting.Lifetime[0]
info: Content root path: P:\PE III\Personal\Eric\Projects\+HiokiInterfaceExecutable
```



This app is hosted locally, so there's not a website to go to (in the traditional sense). Open any browser and type in the address from the second line of the terminal output. I'm 99% sure it's stable at localhost:5000, but it will always match whatever you see in the terminal created by the .exe.



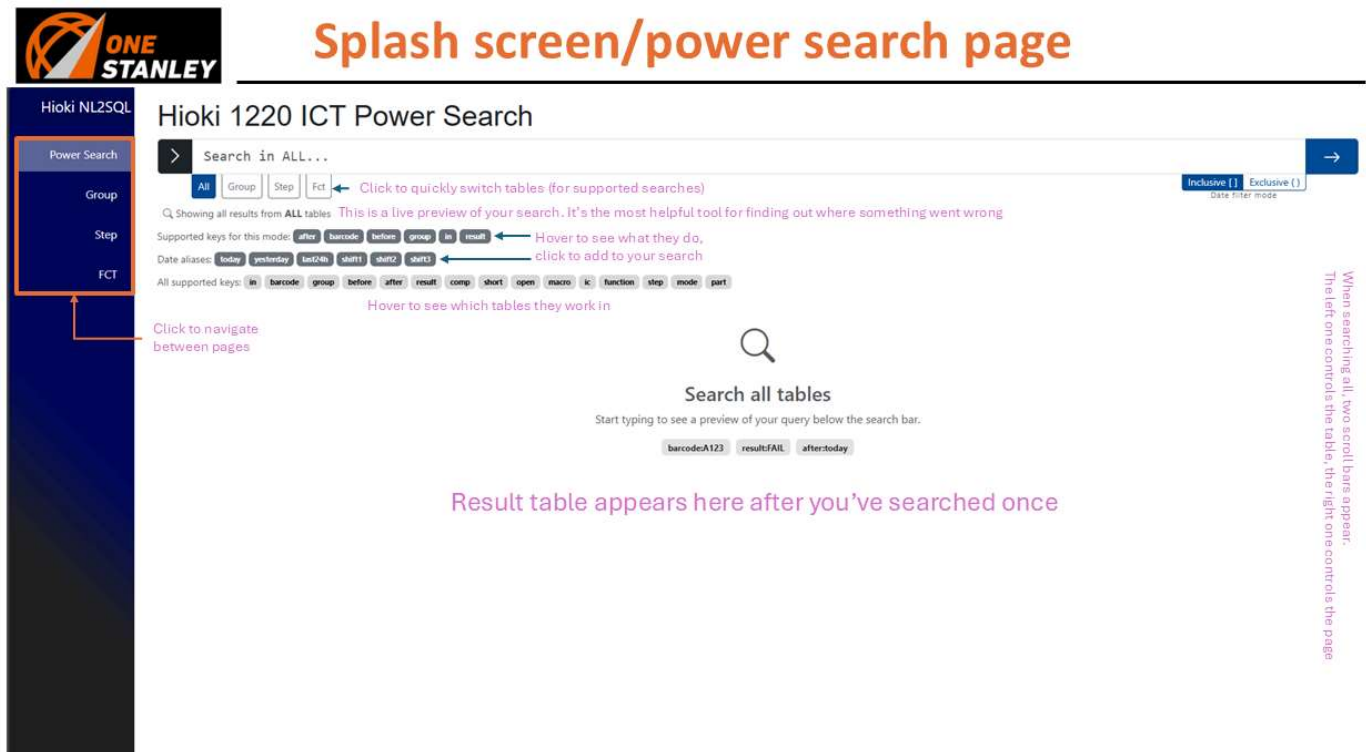
Side note: the app's lifespan is tied to the terminal, not the browser. Thus, it doesn't care when you close the tab, go to a different site, or even close the browser entirely. The only way to end it is to go back to the terminal and either close it or press Ctrl+C as it indicates. Once the app closes, all tabs currently viewing it will try to reconnect a couple times before giving up.

The left hand side of the page has tabs for navigation between the four pages: Group, Step, FCT, and Power Search. The group, step, and FCT pages are what are known as visual query builder pages (VQ pages) and provide a simpler interface at the cost of customizability and the "search all" feature. Please allow some loading time after navigating to a page for the first time as the browser loads the necessary table. Subsequent loads should be much faster, as they do not require more

database queries. Refreshing the page un-loads the browser, so you can expect another slow load time after refreshing.

It is recommended to use only the power search page, or only the VQ pages, as the power search page “borrows” the engines from the VQ pages, forcing additional reloads when navigating from VQ to power search or vice versa. Additionally, this means the VQ pages will have their fields automatically filled from the power search page. This is kind of cool (but mostly an artifact of the aforementioned engine borrowing).

The power search page doubles as the home page, so when you first open it, there will be a message instructing you to use the search bar and keep an eye on the live preview. This message will be replaced by the result table after you’ve completed a search. This is not the case for the VQ pages, which automatically load the table when they are opened.



Power Search

*Terminology note: a ‘tag’ is any key-value pair usable in the search bar, while a ‘filter’ is the key from a tag that is not **in***

The power search page consists of a search bar with table tabs, a preview, and interactable tags supported by the search bar (plus the result table itself). Any tag in dark gray can be clicked to be added to the search, and any tag can be hovered over to see a helpful tooltip containing details

about what the tag does and which tables in which it is applicable. Clicking a tag will turn it blue to denote its presence in the search, and an X will appear for removing the tag.

Once you're ready, use space-separated key:value syntax (e.g. `in:group comp:pass`) to specify your filters.

The **in** tag is particularly special, because it determines which table(s) to search, and by extension, which filters are available. Without specifying an **in** tag, you search all tables, which can be helpful for a few cases, like tracking a barcode (`barcode:targ3txxbarc0d3`) or getting all non-passing test results during a certain shift (`-result:pass before:shift1 after:shift1` in inclusive mode), but your filters are limited to just the six core ones you see under "Supported keys in this mode". To gain access to a filter, your **in** tag must target a table in which that filter has a column. If you're wondering which table(s) supports a specific filter, hover over the filter where it's listed in "All supported keys". If you're wondering what filters are supported by a specific table, just type `in:<table name>` and take a look at the newly populated key list in "Supported keys for this mode". For example, `step:3` will fail because the step filter is not supported by the group table (so all cannot be searched), but either `in:step step:3` or `in:fct step:3` works because both of those tables support the **step** tag

Filters can filter *by* (default, find entries where values match what you specify) or filter *out* (find entries where the value you specify is absent) by appending a hyphen to the beginning of your key (e.g. `-group:1` finds every result belonging to a group that's not 1). This works for all tags except the **in** tag (because it determines membership, not details) and the before/after tags (because `-before` is equivalent to `after`, and vice versa). Using a nega-filter will turn it red in the tag list. This does not mean it is wrong, it is just a visual indicator that the filter is negated because hyphens are tiny.

Values can be wrapped in quotes (specifically double quotes, `"`) in order to "escape" the special characters space and colon `:`. You're welcome to use quotes for any value, as they're just a grouping construct and will not affect the value you specify, but they are necessary when specifying a value for a filter which includes a space (e.g. `date+time`, some test modes) or a colon (e.g. `time`). If at any time there comes a point where it would be sufficiently advantageous to use a literal quote in a value, the regular expression in the parser could be amended to accept a different grouping

construct (e.g. parentheses or other brackets) without much difficulty.



Using the search bar

Hioki NL25QL

Power Search

Group

Step

FCT

Hioki 1220 ICT Power Search

Use space-separated key:value syntax. Put a hyphen at the start of any key to filter out the value.
Wrap a value in quotes if you wish to include a space or colon (including time) inside the value.

>

in:fct before:"today 11:23" -mode:"flash rom" after:"lastweek shift2"

X

→

All

Group

Step

FCT

This is what the tool thinks you said. It is always right.

Q Searching FCT where DATE is BEFORE '2026-02-26 11:23:00' and MODE does NOT contain 'flash rom' and DATE is AFTER '2026-02-19 15:30:00'

Supported keys for this model: after, barcode, before, group, in, mode, result, step

As you type, filters will populate based on your 'in' tag, and change color based on presence

Date aliases: today, yesterday, last4hrs, shift1, shift2, shift3

All supported keys: in, barcode, group, before, after, result, comp, short, macro, k, function, step, mode, part

Gray: filter is not in search
Blue: filter is in search
Red: negative filter in search

These do not apply for date aliases

With these settings, the date range is from the start of 2nd shift on 2/19 to 11:23 on 2/26 (see preview)

FCT Results

The table becomes faded if you haven't pressed enter yet to remind you that it's waiting for you to search. You can't interact when it's like this, but you can refresh if you want to go back to the search that generated this table

Showing 1 to 50 of 1550 Rows: 50 [Save to CSV](#)

Go to: 1 Go of 31 [Previous](#) [Next](#)

BARCODE	TIME	GROUP	STEP	RESULT	MEASUREMENT GROUP	STEP COMMENT	PART POSITION	TEST MODE	HIGH-PIN	LOW-PIN	REFERENCE VALUE	MEASURED VALUE
43296000020A25xx1558825120816072786	12/12/2025 2:00:09 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.979
43296000020A25xx1558825120816072786	12/12/2025 2:00:09 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
43296000020A25xx1558825120816072782	12/12/2025 1:59:44 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.979
43296000020A25xx1558825120816072782	12/12/2025 1:59:44 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
43296000020A25xx1558825120816082814	12/12/2025 1:59:19 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.979
43296000020A25xx1558825120816082814	12/12/2025 1:59:19 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
43296000020A25xx1558825120816072785	12/12/2025 1:58:54 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.972
43296000020A25xx1558825120816072785	12/12/2025 1:58:54 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
43296000020A25xx1558825120816072789	12/12/2025 1:58:23 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.971
43296000020A25xx1558825120816072789	12/12/2025 1:58:23 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
43296000020A25xx1558825120816072788	12/12/2025 1:57:54 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.977
43296000020A25xx1558825120816072788	12/12/2025 1:57:54 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144

Upon editing the search bar after executing a search, the table will fade to denote “staleness”. This removes its interactability until the next search is executed or the page is refreshed. Sometimes the table will appear stale even though it is not. I have no idea why this happens. Fortunately, a refresh or two will always fix this.

Below is the table used across all four pages. It (ideally) behaves exactly the same way on each one. The most interesting feature that you won't find in similar search tools is what I refer to as the barcode “drill-down” Suppose you find a particularly interesting test result and you wish to learn more about that specific part. You can click on the barcode in question and you'll be sent to the power search page (if you're not already there, as this works on all four pages) with a query preloaded to search all for that barcode. This will show you all of the results (group, step, and FCT) associated with that barcode.

For wider tables (step and FCT), use the horizontal scrollbar beneath the table or `Shift+scroll` while hovering over it to see the columns that are off-screen.

Hioki NL2SQL
Power Search
Group
Step
FCT

Hioki 1220 ICT Power Search

Group Results

The results you're seeing now

Results per page

Save the entire search's results to a CSV file (check your browser's download settings)

Jump to a specific page

Go to: 1 of 18

Showing 1 to 50 of 864 Rows: 50

BARCODE	TIME	GROUP	# OF TIMES TESTED	ALLRESULT	COMPONENTTEST	SHORTTEST	OPENTEST	ICTEST	MACROTEST	FUNCTIONTEST
432960000020A25xc1558825120816072706	12/12/2025 2:00:09 PM	1							UN-T	PASS
432960000020A25xc1558825120816072702	12/12/2025 1:59:44 PM	1							UN-T	PASS
432960000020A25xc1558825120816082814	12/12/2025 1:59:19 PM	1	31137	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072705			36	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072709			35	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072708	12/12/2025 1:57:54 PM	1	31134	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072712	12/12/2025 1:57:28 PM	1	31133	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072711	12/12/2025 1:56:58 PM	1	31132	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072701	12/12/2025 1:56:32 PM	1	31131	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072704	12/12/2025 1:55:58 PM	1	31130	PASS	PASS	UN-T	UN-T	UN-T	UN-T	PASS
432960000020A25xc1558825120816072707	12/12/2025 1:55:07 PM	1	31129	FAIL	FAIL	UN-T	UN-T	UN-T	UN-T	UN-T
432960000020A25xc1558825120816072710	12/12/2025 1:54:33 PM	1	31128	FAIL	FAIL	UN-T	UN-T	UN-T	UN-T	UN-T

Dates and times (particularly shift aliases) almost require a separate guide of their own, because humans have a very strange perception of time. I will do my best to explain, but hopefully you will find this intuitive. To begin with, clicking a date alias adds it to the search. If there is a key awaiting a value at that time, only the alias will be appended. Otherwise, the tool assumes the after tag and appends after:<alias>. Executing the search replaces all aliases with their interpreted value, increasing customizability in case of shifts on overtime, etc.

Strictly speaking, the tool works best with full ISO datetimes in 24-hour time, but those get frustrating fast, so there is support for simplified dates (e.g. MM-DD, MM/D, MM/D-yy) and times (e.g. 1:04PM, but careful not to add a space between the time and A/PM or the tool doesn't recognize the time at all, even with quotes). All this goes to say, use the shorthand that works for you, but if you suspect the tool isn't understanding you, try ISO's "YYYY-MM-DD HH:mm:ss" (don't forget the quotes)

Here are the rules for how alias substitution is applied (I'll cover how endpoints are resolved next):

- If a time (shift) alias is used without an accompanying date part, the app infers that you meant the most recent time that shift ran, including the current shift. For example, running a search on 2/25 at 10:47 would resolve:
 - shift1: 2/25, 7:00 to 15:29
 - shift2: 2/24, 15:30 to 22:29
 - shift3: 2/24, 22:30 to 2/25, 6:59

- If a time (not alias) is used without an accompanying date part, the app infers that you meant the most recent date that time occurred (as to avoid searching the future, which would certainly yield no results).
 - This means if that time has not yet occurred today, it will refer to yesterday at that time (e.g. after:11:00 on 2/25 at 10:47 will search after 11:00 on 2/24).
 - As previously mentioned, this **must** be wrapped in quotes, otherwise the tool will not recognize it as a time but as a key (bad).
 - Semantically, this is a little unreliable, so if you want to specify a specific time, I'd recommend putting it with a date (below).
- If a date (or date alias) is used without an accompanying time part, the app infers that you want to cover the whole day. This is the most intuitive one.
- If a date is used with a time (either may or may not be an alias), all is well, as long as you put the date *before* the time, and the whole value is wrapped in quotes.
 - Please don't put the time before the date. You won't get an error, but the app will lose its mind. It should hopefully feel normal to specify time after date.



Date aliases: substitution

Executing the search will substitute date aliases with their actual values

Hioki 1220 ICT Power Search

> before:shift2 after:yesterday

All Group Step Fct

Inclusive {} Exclusive {}
Date filter mode

Searching ALL where DATE is BEFORE '2026-02-26 22:29:59' and DATE is AFTER '2026-02-26 00:00:00'

Supported keys for this mode: after before barcode group in result

Date aliases: today yesterday last24h shift1 shift2 shift3 When you click an alias, it's added to the search. If you have a key waiting for a value, only the alias is appended. Otherwise, it assumes "after: <alias>"

All supported keys: in barcode group before after result comp short open macro ic function step mode part

If you use a shift alias without a date, the app infers that you meant the most recent time that shift ran (including the current shift)
 If you use a time without a date, the app infers you meant today at that time. This MUST be in quotes, otherwise the tool cannot read it properly.
 24hr time works best, but you can use AM/PM just the same IF you don't put a space between the time and AM/PM (the tool does not understand that)
 If you use a date or date alias without a time, the app infers that you meant that entire day, adjusting midnight/23:59 for after/before, respectively

Hioki 1220 ICT Power Search

> before:"2026-02-26 22:29:59" after:"2026-02-26 00:00:00"

All Group Step Fct

Inclusive {} Exclusive {}
Date filter mode

Searching ALL where DATE is BEFORE '2026-02-26 22:29:59' and DATE is AFTER '2026-02-26 00:00:00'

Supported keys for this mode: after before barcode group in result

Date aliases: today yesterday last24h shift1 shift2 shift3

All supported keys: in barcode group before after result comp short open macro ic function step mode part

If you don't use a before/after tag (I recommend you do once the database fills up) or specify a clear time (not alias), the inclusive/exclusive toggle won't affect your search at all. If you use any alias, or use a date without a time (i.e. you specify what effectively represents a timespan), this little switch next to the clear and execute search buttons will determine whether the time span you specify is included or excluded (non-strict vs. strict inequality, $\leq$ vs. $<$). The rule of thumb is that an **after** tag,

this always gets the beginning in inclusive mode and the end in exclusive mode, but **before** does the opposite: end in inclusive, beginning in exclusive. This should be intuitive.

For shift aliases, the beginning and end are well defined:

- shift1 - 07:00-15:29
- shift2 - 15:30-22:29
- shift3 - 22:30-06:59 (00:00-06:59 is on current day, 22:30-23:59 on previous day)
 - Suppose this is run on 1/10 (different to avoid confusion with the following example)
 - inclusive before:shift3 = all entries before and including shift3 of 1/10: before 1/10 at 06:59
 - exclusive before:shift3 = all entries strictly before shift3 of 1/10: before 1/9 at 22:30
 - inclusive after:shift3 = all entries after and including shift3 of 1/10: after 1/9 at 22:30
 - exclusive after:shift3 = all entries strictly after shift3 of 1/10: after 1/10 at 06:59

For dates, this just resolves to midnight or 11:59 (this applies to any date; here is an example)

- 2/25 - 2/25 00:00-2/25 23:59
 - inclusive before:2/25 = all entries before and including 2/25: before 2/25 at 23:59
 - exclusive before:2/25 = all entries strictly before 2/25: before 2/25 at 00:00
 - inclusive after:2/25 = all entries after and including 2/25: after 2/25 at 00:00
 - exclusive after:2/25 = all entries strictly after 2/25: after 2/25 at 11:59

Of course, this also works with date+shift alias specification (the times from the shift alias, offset by the date).



Date aliases: inclusivity

These searches are identical except for the inclusivity

These searches were performed during first shift on 2/27, so a auto-detection mapped shift1 to the current shift and shift2 to yesterday's 2nd shift. The shift3 alias was not used, but if it were, it would correspond to the 3rd shift in between (spanning 2/26 to 2/27)

Hioki 1220 ICT Power Search

> before:shift1 after:"yesterday shift2"

All Group Step Fct

Q Searching ALL where DATE is BEFORE '2026-02-27 15:29:59' and DATE is AFTER '2026-02-26 15:30:00'

Supported keys for this mode: after before barcode before group in result

Date aliases: today yesterday last24h shift1 shift2 shift3

All supported keys: in barcode group before after result comp short open macro ic function step mode part

This gathers all the entries between the time 2nd shift started yesterday and the time 1st shift will end today

Searching before the current shift with inclusive mode enabled is the only time the tool will search the future (to capture the entire first shift)

Inclusive [] Exclusive ()
Date filter mode

Hioki 1220 ICT Power Search

> before:shift1 after:"yesterday shift2"

All Group Step Fct

Q Searching ALL where DATE is BEFORE '2026-02-27 06:59:59' and DATE is AFTER '2026-02-26 22:30:00'

Supported keys for this mode: after before barcode before group in result

Date aliases: today yesterday last24h shift1 shift2 shift3

All supported keys: in barcode group before after result comp short open macro ic function step mode part

This gathers all the entries between the time 2nd shift ended yesterday and the time 1st shift started today

Inclusive [] Exclusive ()
Date filter mode

Non-fatal errors (warnings) are ones where the app thinks it understood what you meant and can automatically correct it, or that the result of excluding bad input would be close to what you wanted. This includes things like duplicate tags, negation where it shouldn't be, and tags missing a key or value.



Errors: non-fatal

Hopefully, your error list won't be this long. Errors with a yellow background are non-fatal, and the search will execute with even this many, automatically applying the changes it mentions.

Hioki NL2SQL

Power Search

Group

Step

FCT

Hioki 1220 ICT Power Search

- ⚠ Duplicate in tag. This search is now infct.... All previous uses of this key are ignored.
- ⚠ The active in tag cannot be negated. This search is now infct....
- ⚠ Negation should be applied to the key instead of value, but the in tag cannot be negated anyway. This search is now infct. If the messages are taking up too much space, you can click the X on each to dismiss them, but they'll automatically go away on your next search
- ⚠ Tag missingVal: is missing a value. This search excludes missingVal:
- ⚠ The value for before started with a hyphen. This search is now -before:yesterday.... To search for a literal hyphen, use quotes like -before:"-yesterday".
- ⚠ The before tag cannot be negated. This search is now before : yesterday...
- ⚠ The after tag cannot be negated. This search is now after : today...
- ⚠ Your start date is after your end date. This search is now after:2026-02-26 before:2026-02-27...
- ⚠ Unrecognized filter without key: missingKey. This search excludes missingKey.

> in:drop -in:-fct missingVal: -before:"2026-02-27 00:00:00" -after:"2026-02-26 23:59:59" missingKey

All Group Step Fct

Q Searching FCT where DATE is BEFORE '2026-02-27 00:00:00' and DATE is AFTER '2026-02-26 23:59:59'

Supported keys for this mode: after before barcode before group in macro result step

Date aliases: today yesterday last24h shift1 shift2 shift3

All supported keys: in barcode group before after result comp short open macro ic function step mode part

Side note: this is alias translation at work. Observe how the error messages refer to 'today' and 'yesterday'

Inclusive [] Exclusive ()
Date filter mode

This particular set of amended filters didn't yield results, so they are not shown

Fatal errors are ones where you messed up in a way that the app can't automatically correct, and simply excluding the bad input would likely yield different results from what you wanted. This includes things like invalid target for the **in** tag, a mismatched tag for a table, and incorrect data type for a filter.



Errors: fatal

With just one **red** background error, your search won't even execute. These usually mean the tool had no idea what you meant and needs your help to correct it.

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Power Search

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1 flibber is not a valid table. The **in** keyword only accepts the values *all*, *group*, *step*, or *fct*.

2 Security Issue: The value for **macro** contains forbidden keywords.

3 The tag **comp** is not available when searching **fct**. Try a different tag or search a table with that attribute.

4 The tag **flobber** wasn't recognized. Try using the table and key options below the search bar.

5 Invalid value for the **step** tag. **every** is not a whole number.

6 Invalid value for the **before** tag. **eleventeen** (read as 1/1/0001 12:00:00 AM) is not a valid date or alias. Please use "YYYY-MM-DD HH:mm:ss" (ISO formatting) or a shortcut below.

7 Invalid value for the **after** tag. **2030-12-12** is in the future. Did you mean to search before that date?

>

in:flibber macro:"alter table" comp:good flobber:true step:every before:eleventeen after:"2030-12-12"

×

→

AllGroupStepFCT

Inclusive [] Exclusive ()
Date filter mode

Showing all results from FCT

Supported keys for this mode: after > barcode before > group in > mode result step >

Date aliases: today yesterday last24h shift1 shift2 shift3

All supported keys: in barcode group before after result comp short open macro K function stop mode part

We'll discuss this symbol on the next slide



This is what it looks like when your search doesn't yield any results because your filters were too strict. Unless you get errors in addition to this screen, your syntax was fine, but nothing in the database matches your search criteria. There's a chance you trigger this one by simply searching

the wrong table.



No results

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Power Search

Group

Step

FCT

Hioki 1220 ICT Power Search

> in:all group:1000000000

All Group Step Fct

Searching ALL where GROUP is "1000000000"

Supported keys for this mode: after barcode before group in result

Date aliases: today yesterday last24h shift1 shift2 shift3

All supported keys: in barcode group before after result comp short open macro k function step mode part

Inclusive [] Exclusive []
Date filter mode



No results matched these criteria.
Try adjusting your filters or checking for typos.

There weren't any errors in this search, but there's no group with a number that high, so there weren't any results. The tool can't really help you here, so just double check the preview to make sure your search is what you really meant.

For every successful power search (i.e. no fatal errors), the page's URL will update to reflect the filter, sort, and page information. This way, if you find search results you'd like to share with someone else, you can copy the URL from the address bar and send it to someone else with the program. If the program is running on their machine, they can just paste it into their address bar and immediately see the results of your search. This is also a great way to "save" a favorite search that you like to run every now and then. In either case, just remember the page number will change over

time as more entries are added to the database.



Page # and size (p and ps parameter)

Sharing/saving a search

The URL updates on a successful search, sort, or page change, so it always matches the search that was used to get the table being displayed

Filters (q parameter)

Sort column name and direction (s and d parameter)

Hioki NL2SQL

Power Search

Group

Step

FCT

in:step mode:R-CC step:9

Inclusive (1) Exclusive (1)

Searching STEP where MODE contains 'R-CC' and STEP is '9'

Supported keys for this model: after, barcode, before, group, in, mode, part, remain, step

Date aliases: today, yesterday, last4th, shift1, shift2, shift3

All supported keys: in, barcode, group, before, after, result, comp, short, open, macro, k, function, step, mode, part

This is also a good way to "bookmark" a search you like to execute from time to time

The power search page builds a URL from your search so it can be easily shared. Simply copy the search bar and have someone else paste it after they start the program on their machine. Barring no changes in the database, this will show them the exact same results, down to the filters, sorts, and page.

Step Results

Showing 26 to 50 of 238 Rows: 25 Save to CSV

Go to: 2 of 10 Previous Next

PART NAME	HIGH-PIN	LOW-PIN	PART POSITION	MODE	MEASUREMENT RANGE	HIGH LIMIT	LOW LIMIT	UNIT	ACTUAL (MOUNTED) VALUE	REFERENCE VALUE	MEASURED VALUE
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.697	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.698	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.698	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.698	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.699	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.699	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.699	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.7	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.7	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.7	
R8	5	4	A-7	R-CC	5	12	7	39	9.559	9.7	

VQ Pages

The VQ pages are exactly the same as the power search page under the hood, just with an interface that's easier to learn. If you don't care about having absolute control over the fine tuning of your filters, this is a great way to pick up the app and get the results you want. The two biggest limitations of the VQ pages are the inability to search all tables and the inability to specify a time (outside of shift aliases), so if that's not a problem, this is for you! The table operates exactly the

same as it does on the power search page, just without an updating URL.



The visual query builder pages

- ❖ The visual query builder (VQ) pages are the app's three other pages that provide a simpler interface than the search bar at the cost of customization
- ❖ They are all mostly the same, but provide different form fields depending on what filters are available for their respective tables
- ❖ Table controls (i.e., sorts, pagination, save to CSV) are the [same as the power search page](#)
- ❖ Because each VQ page only touches one table, it's loaded immediately when you navigate here





Group table VQ page



Hioki 1220 ICT Group Test Results

Lower date granularity: only shift options for times Click mm/dd/yyyy to type, or the calendar icon to choose from a calendar display

BARCODE: START DATE: All Shifts: END DATE: All Shifts: GROUP #: OVERALL RESULT:

COMPONENT TEST RESULT: SHORT TEST RESULT: OPEN TEST RESULT: MACRO TEST RESULT: IC TEST RESULT: FUNCTION TEST RESULT:

Each dropdown contains only the result types present in the database

Still only applies for date filters

Inclusive ☐ Exclusive ☐ Clear Execute

Date filter mode

Showing all results from GROUP Automatically scrolls down to "fullscreen" the table

Showing 1 to 50 of 1760 Rows: 50 Save to CSV Go to: 1 of 36 Previous Next

BARCODE	TIME	GROUP	# OF TIMES TESTED	ALLRESULT	COMPONENTTEST	SHORTTEST	OPENTEST	ICTEST	MACROTEST	FUNCTIONTEST
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	250	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	2	250	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	3	250	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	4	250	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418373200	2/9/2026 2:42:14 PM	1	249	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418373200	2/9/2026 2:42:14 PM	2	249	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418373200	2/9/2026 2:42:14 PM	3	249	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418373200	2/9/2026 2:42:14 PM	4	249	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418378400	2/9/2026 2:41:20 PM	1	248	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418378400	2/9/2026 2:41:20 PM	2	248	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418378400	2/9/2026 2:41:20 PM	3	248	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T
94362016001E00000EL21826020418378400	2/9/2026 2:41:20 PM	4	248	PASS	PASS	PASS	PASS	UN-T	UN-T	UN-T



Step table VQ page



Hioki 1220 ICT Step Test Results

The tool searches for whether a barcode contains this filter, so you can enter any portion of it here

Like the power search page, you can enter a date, time, or both. The same inference rules apply

BARCODE: START DATE: All Shifts: END DATE: All Shifts: GROUP #: STEP #:

PART NAME: TEST MODE: RESULT:

The clear button removes your filters AND resets the table. If the app is acting up, try this, then a refresh if it's still no good

Inclusive ☐ Exclusive ☐ Clear Execute

Date filter mode

Showing all results from STEP

Showing 1 to 50 of 54754 Rows: 50

Save to CSV

 Go to: 1 of 1096

Previous Next

BARCODE	TIME	GROUP	STEP	TEST TIMES	RESULT	PART NAME	HIGH-PIN	LOW-PIN	PART POSITION	MODE	MEASUREMENT RANGE	HIGH LIMIT	LOW LIMIT	UNIT
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	2	250	PASS	R1	6	5	A-7	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	3	250	PASS	R2	6	5	A-7	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	4	250	PASS	R3	6	5	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	5	250	PASS	R4	6	5	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	6	250	PASS	R5	6	5	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	7	250	PASS	R6	6	5	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	8	250	PASS	R7	5	4	A-7	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	9	250	PASS	R8	5	4	A-7	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	10	250	PASS	R9	5	4	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	11	250	PASS	R10	5	4	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	12	250	PASS	R11	5	4	A-6	R-CC	5	12	7	
94362016001E00000EL21826020418375900	2/9/2026 2:43:09 PM	1	13	250	PASS	R12	5	4	A-6	R-CC	5	12	7	



FCT table VQ page

Hioki NL2SQL

Power Search

Group

Step

FCT

Hioki 1220 ICT Functional Test Results

BARCODE	START DATE	END DATE	GROUP #	STEP #	TEST MODE
Scan or type...	mm/dd/yyyy 📅 All Shifts	mm/dd/yyyy 📅 All Shifts			All Modes
RESULT					
All Results					

Like the other dropdowns, the test mode filter contains only the entries present in the database

The FCT page looks a lot like the step page, but without part name. There's lots of columns for this table, so use the horizontal scrollbar or Shift+scroll while hovering over the table to see them all.

Date filter mode

Showing all results from FCT

Showing 1 to 50 of 1550 Rows: 50

Go to: 1 of 31

BARCODE	TIME	GROUP	STEP	RESULT	MEASUREMENT GROUP	STEP COMMENT	PART POSITION	TEST MODE	HIGH-PIN	LOW-PIN	REFERENCE VALUE	MEASURED VALUE
432960000020A25xc1558825120816072706	12/12/2025 2:00:09 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.979
432960000020A25xc1558825120816072706	12/12/2025 2:00:09 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
432960000020A25xc1558825120816072702	12/12/2025 1:59:44 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.979
432960000020A25xc1558825120816072702	12/12/2025 1:59:44 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
432960000020A25xc1558825120816082814	12/12/2025 1:59:19 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.979
432960000020A25xc1558825120816082814	12/12/2025 1:59:19 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
432960000020A25xc1558825120816072705	12/12/2025 1:58:54 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.972
432960000020A25xc1558825120816072705	12/12/2025 1:58:54 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
432960000020A25xc1558825120816072709	12/12/2025 1:58:23 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.971
432960000020A25xc1558825120816072709	12/12/2025 1:58:23 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144
432960000020A25xc1558825120816072708	12/12/2025 1:57:54 PM	1	5	PASS	1	VOLT CHECK	A-0	VoltDC	130	129	5	4.977
432960000020A25xc1558825120816072708	12/12/2025 1:57:54 PM	1	8	PASS	1	30AA ROM	A-0	FLASH ROM	4	30AA_HS_105	32144	32144